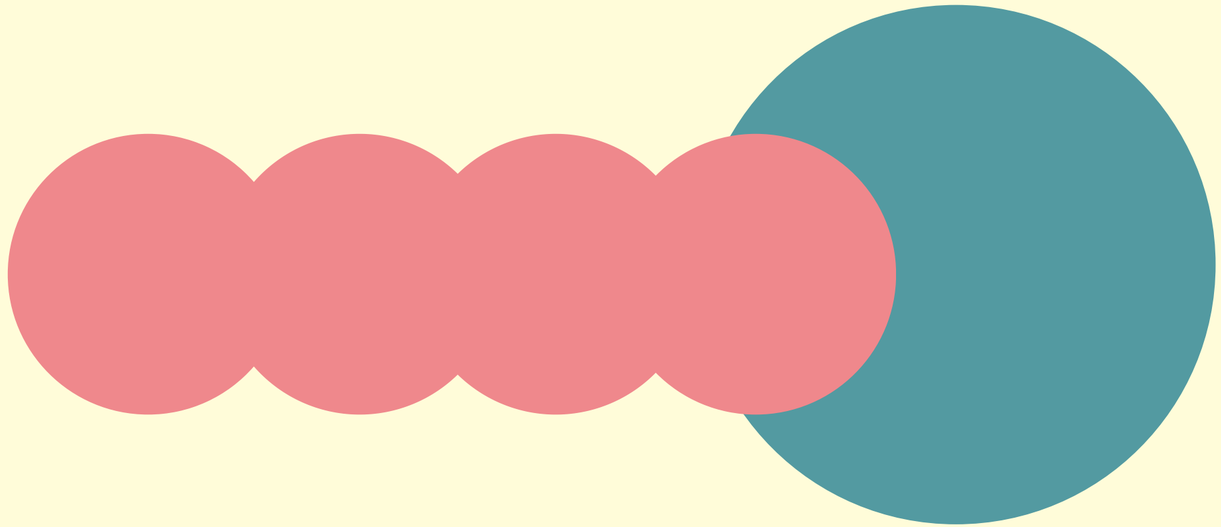




THE CITY AS SYSTEM:

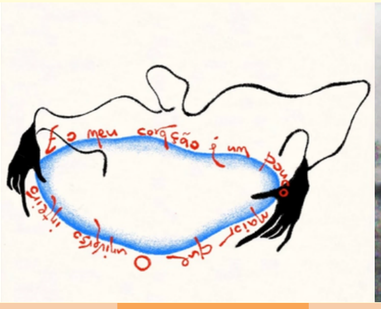
CODE, MEMORY, AND THE AESTHETICS OF URBAN ENCOUNTER

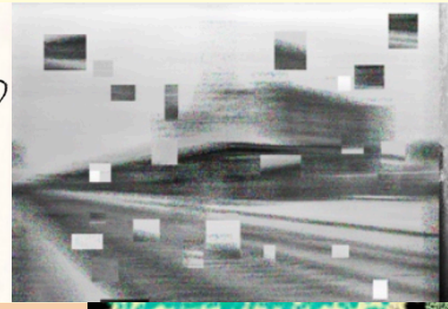
Yi Ru (Raya)

Final Portfolio · COMPSCI 208

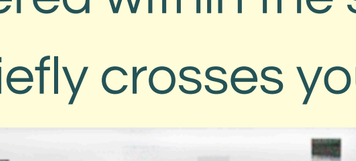
My computational practice begins with a city and ends with a question about presence. Across six projects, I kept returning to urban space—not to document it, but to ask what a city feels like from the inside: what it sounds like to someone who grew up on its metro lines, what it looks like when grief and rebuilding are layered within the same photograph, what it means when a stranger's path briefly crosses yours on a Friday night...

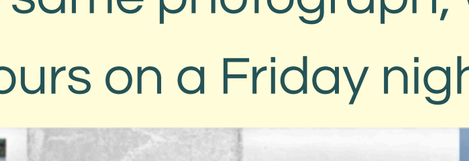


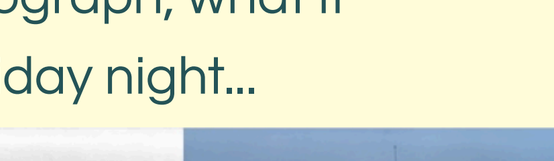


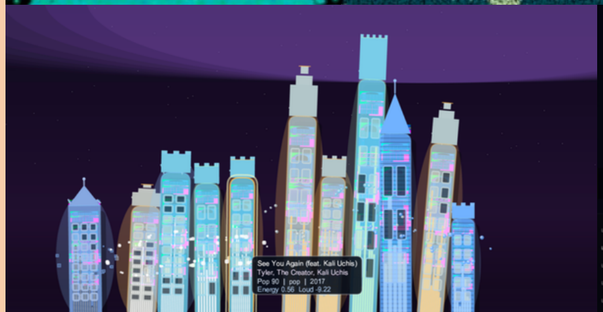


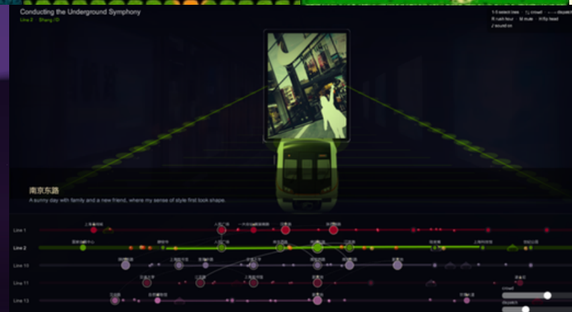








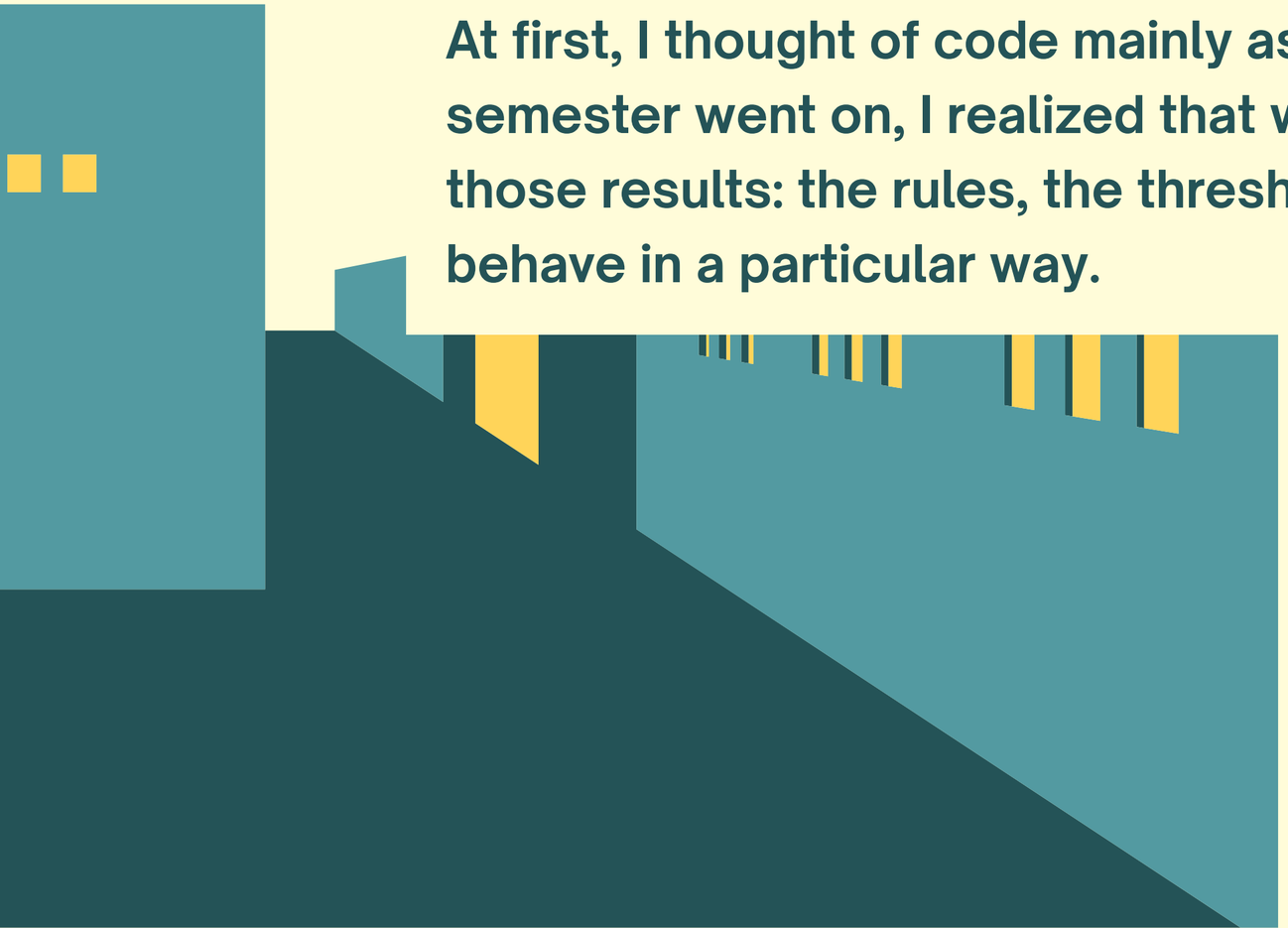




Project1:
When We Meet
Project2:
The Universe Beneath the Grid
Project3:
Sliding Through A Hundred-Year Train
Project4:
City Through Four Lenses
Project5:
Urban Groove Skyline
Project6:
Conducting the Underground Symphony

Code became the medium through which I could hold all of these things simultaneously — not by illustrating them, but by building systems that enact them. Each project set rules into motion and then stepped back. The work that emerged was always partly mine and partly the machine's, and that collaboration, I discovered, is itself the subject.

At first, I thought of code mainly as a way to generate visual results. But as the semester went on, I realized that what mattered more was the structure behind those results: the rules, the thresholds, the delays, the things that made a system behave in a particular way.

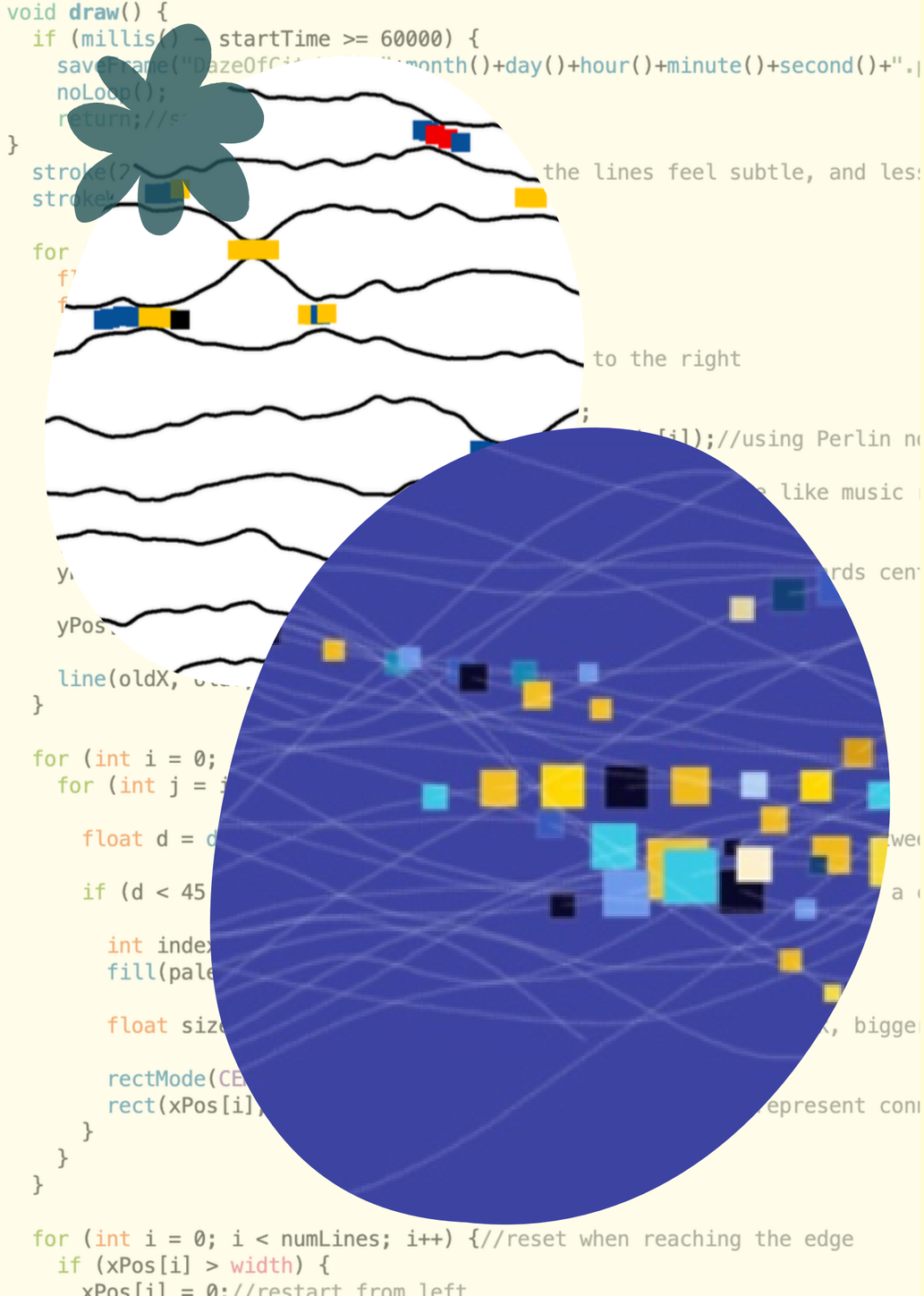


I. RULE AS AUTHORSHIP

Sol LeWitt argues in "Paragraphs on Conceptual Art" that the idea is the machine — execution is secondary. Frieder Nake counters that the computer operates entirely within a system of human design, never escaping the grammar its author wrote.

My first two projects live inside this tension, though they resolve it differently.

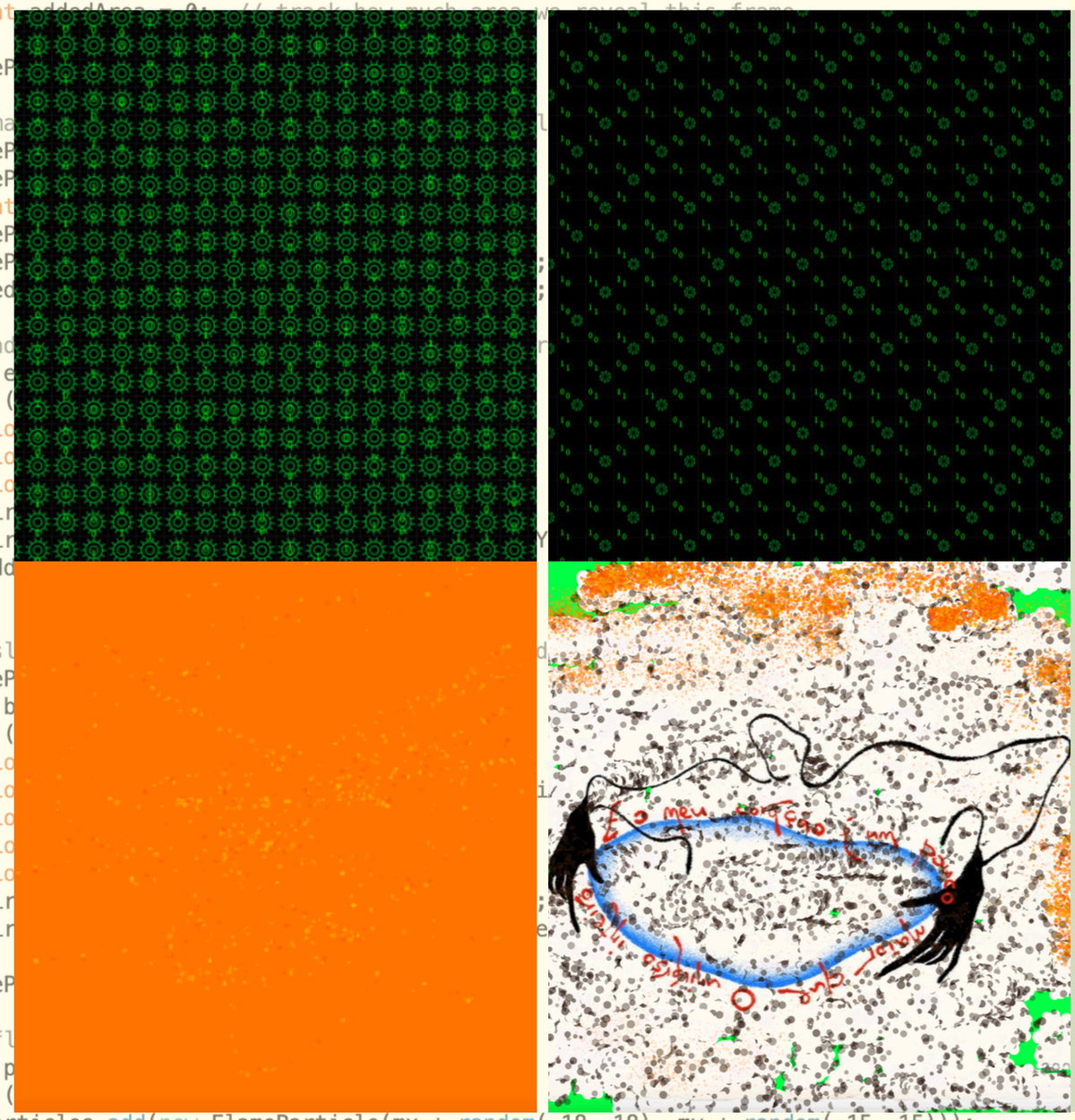
LeWitt, S., 'Paragraphs on Conceptual Art' (1967)
Nake, F., 'There Should Be No Computer Art' (1971)



In *When We Meet*, I wrote rules for movement and proximity, but I did not compose a single intersection. Four semi-transparent paths drift across a nocturnal canvas, pulled gently toward one another by a centripetal force, and wherever two paths come within a threshold distance, a colored square appears — its size and hue determined entirely by the system. What the code produced was something I could not have composed manually: a minute-long performance of near-misses and encounters that is different every time, like the real meetings it models. The program runs for sixty seconds, then saves a PNG and stops. That limit is not a technical constraint — it is an argument. Brevity is the condition of the encounter.

What I learned from this project was that generative art does not become meaningful just because it is random.

In the first versions, the movement felt too loose and the intersections were too sparse, so the work did not yet feel like an encounter. I kept adjusting the speed, opacity, and density until I found a balance where the briefness of the connection became visible. That process made me understand that restraint is not the opposite of expression; sometimes it is what gives expression its shape.

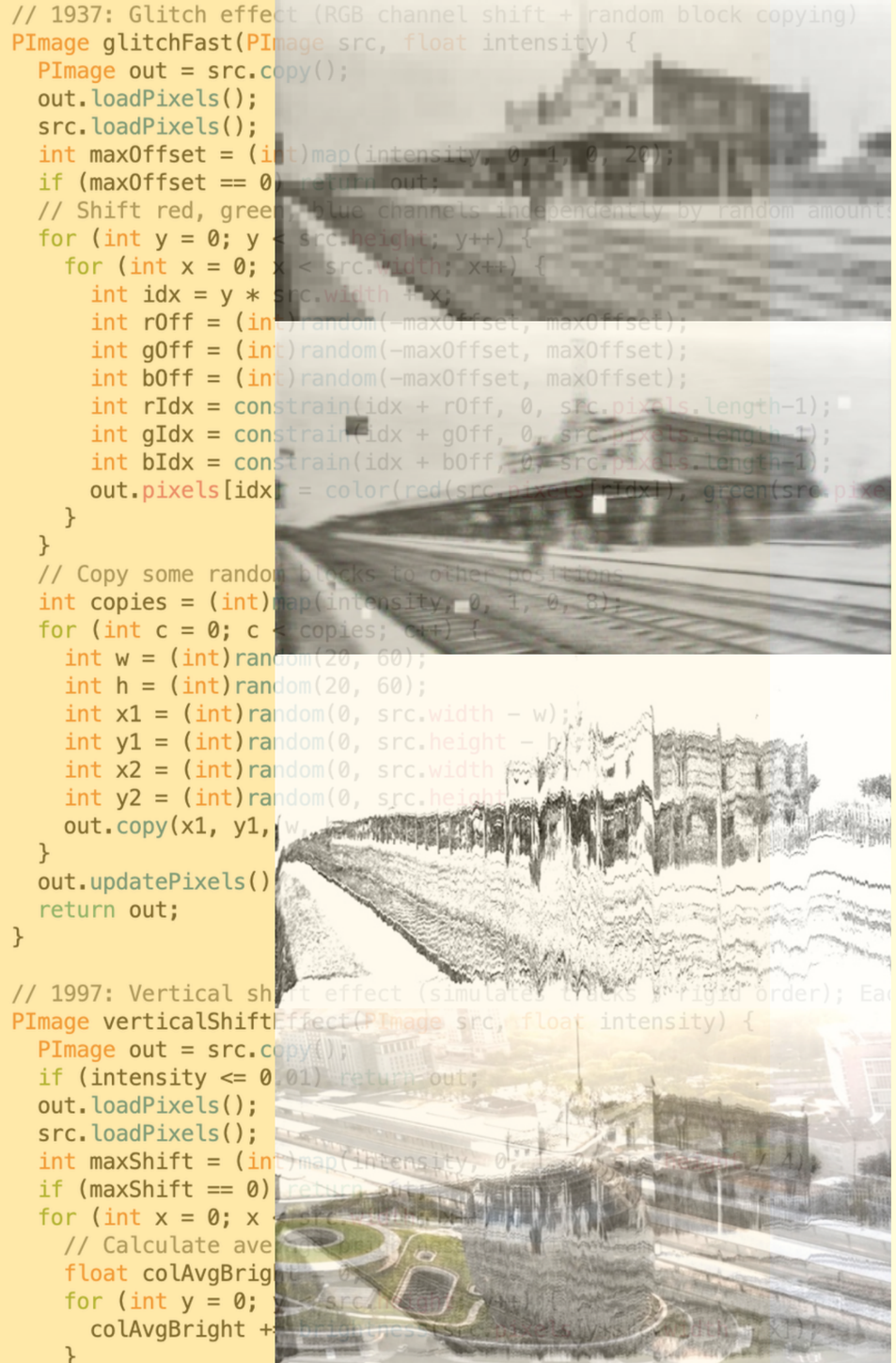


The Universe Beneath the Grid pushes the question of authorship further by staging a confrontation between two modes of control. Mode One is a rigid grid of green binary digits and rotating gears — a system that enforces repetition, suppresses individual deviation, and offers only the cold logic of mechanical production. Mode Two inverts this entirely: the user is handed the ability to "burn" a green overlay and uncover a handwritten poem by Pessoa underneath. Inspired by Marina Abramović's Rhythm 0, in which an audience is granted total agency over the artist's body, the burning interaction is simultaneously creative and destructive — you reveal the poem only by annihilating the system. The rule here is not what to draw, but what must be removed. At first, I treated the two modes as a simple contrast between order and freedom, but that felt too flat. What became more interesting was the fact that the second mode only appears through destruction. I realized that the user is not just revealing an image; they are participating in the work by changing the system itself. Together, these two projects argue that authorship in generative art is always negotiated: sometimes with the system, sometimes against it, and always with a grammar that precedes and exceeds the individual gesture.

II. TRANSFORMATION AS ARGUMENT

John Maeda writes in Design by Numbers that every parameter must have a reason to exist. Golan Levin extends this: the algorithm is secondary — what matters is the relationship between the viewer and transformation.

Maeda, J., Design by Numbers (1999)
Levin, G., 'Some Notes on Visualization Without Computers' (2006)
Boden, M.A. & Edmonds, E.A., 'What Is Generative Art?' (2009)



The two ideas converged when I turned a historical photograph of Jiaxing Railway Station into a temporal argument. Four pixel states map to four historical moments: pixelation for 1907 and the fog of early memory; RGB channel glitch and violent block-copying for 1937 and the Japanese bombing; vertical displacement for the rigid reconstruction of the 1990s; and a gradual fade into contemporary photography for the 2021 forest station. What appears to be a set of visual effects is in fact a claim about how history inscribes itself on a place. A slider replaces linear animation, making the viewer a time traveler rather than a spectator — an act of design that acknowledges, following Levin, that computation should belong to people, not perform for them. One thing I noticed while building this project was that transformation only becomes meaningful when every stage has a clear reason to exist. Early on, some of the transitions looked visually interesting but conceptually weak, so I had to keep asking myself what each effect was actually saying. Once I connected pixelation, glitch, line distortion, and fade to different historical moments, the project stopped feeling like a collection of effects and started feeling like a way of thinking about memory and time.

City Through Four Lenses pursues a different register of transformation: not historical but perceptual. A single city photograph is rendered as approximately 2,500 particles, each retaining its original pixel's color and position as a "home." Mouse movement across the canvas continuously modulates four emotional parameters: Stress, Energy, Order, Warmth, displacing particles away from or returning them toward that home. Moving the cursor toward the Stress corner produces cold blue outward streaks; approaching Warmth casts soft orange light across the image. The city is constant; the lens belongs to the viewer. Boden and Edmonds define generative art as *a system set into motion with some degree of autonomy*, and this definition applies here with precision: I designed the grammar of the four dimensions and their visual consequences, but I cannot predict where 2,500 particles will be after thirty seconds of movement. The specificity of each output is the system's, not mine. What remains distinctively human is the choice of dimensions rather than abstract numerical axes — the curation of the source image, and the final selection of sixteen outputs to submit. These are acts of framing and interpretation that no algorithm performs.

I also learned that complexity is not always the same as depth. My earlier MBTI idea had more categories, but it felt less readable and less honest. Simplifying the project into four emotional dimensions made the system stronger, because the mappings became easier to feel. What I found most interesting was that the code did not just respond to my design; it pushed me to refine the design itself.

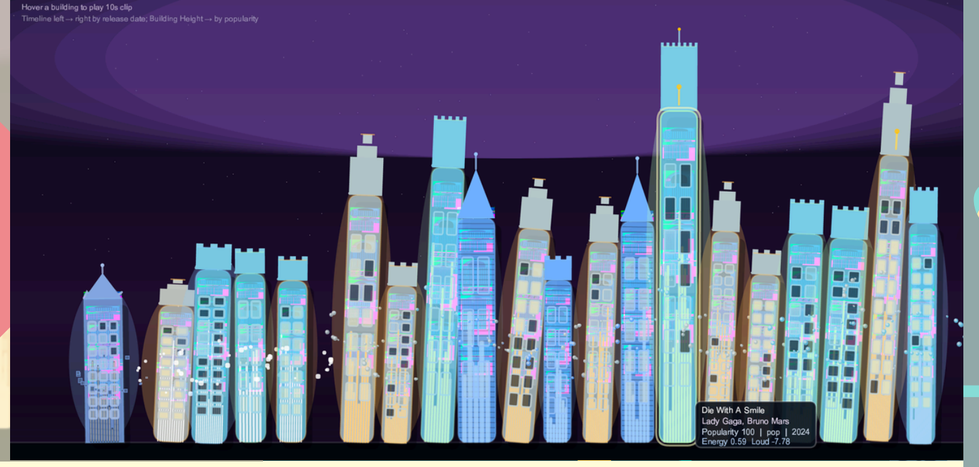
III. DATA AS EMOTIONAL INFRASTRUCTURE

Harold Cohen asks whether a system that evaluates, decides, and satisfies constraints is, in some limited sense, thinking.

I encountered this question most acutely in my final two projects, both of which transform quantitative data into something that feels inhabited.

Cohen, H., 'The Further Exploits of AARON, Painter' (1995)





In *Urban Groove Skyline*, a Spotify dataset of twenty tracks becomes a nocturnal city: popularity controls building height, duration controls width, valence determines roof personality and glow color, and energy drives window flicker and particle activity. The code does not know what valence means emotionally. Yet when release date orders the skyline chronologically and loudness brightens the windows of certain buildings more than others, the system generates a legible visual claim about how mood arranges itself over time. Like AARON, the system lacks intention but not interpretability. It does not argue but it consistently produces something that reads as an argument, which raises the question of where meaning actually resides: in the designer's mapping rules, in the machine's execution, or in the viewer's recognition.

I initially worried that data visualization would make the work feel too cold, but the process showed me that data can carry atmosphere if the mappings are chosen carefully. I had to keep adjusting the balance between clarity and texture, because the first versions felt too much like a chart. The final skyline worked better once I accepted that the goal was not to explain the dataset completely, but to let it become a place that could be felt.

```
void drawTextureAccents(float left, float
  if (!useTexture || genreTexture == null

  pushStyle();
  blendMode(SCREEN);
  tint(255, 50);

  float bandH = h * 0.08;
  image(genreTexture, left, yTop + h * 0.

  float capH = h * 0.05;
  image(genreTexture, left, yTop - capH,

  noTint();
  blendMode(BLEND);
  popStyle();
}

void drawSpectrumBars(float left, float y
  float energyN = norm(energy, minEnergy,
  float loudN = norm(loudness, minLoud, m

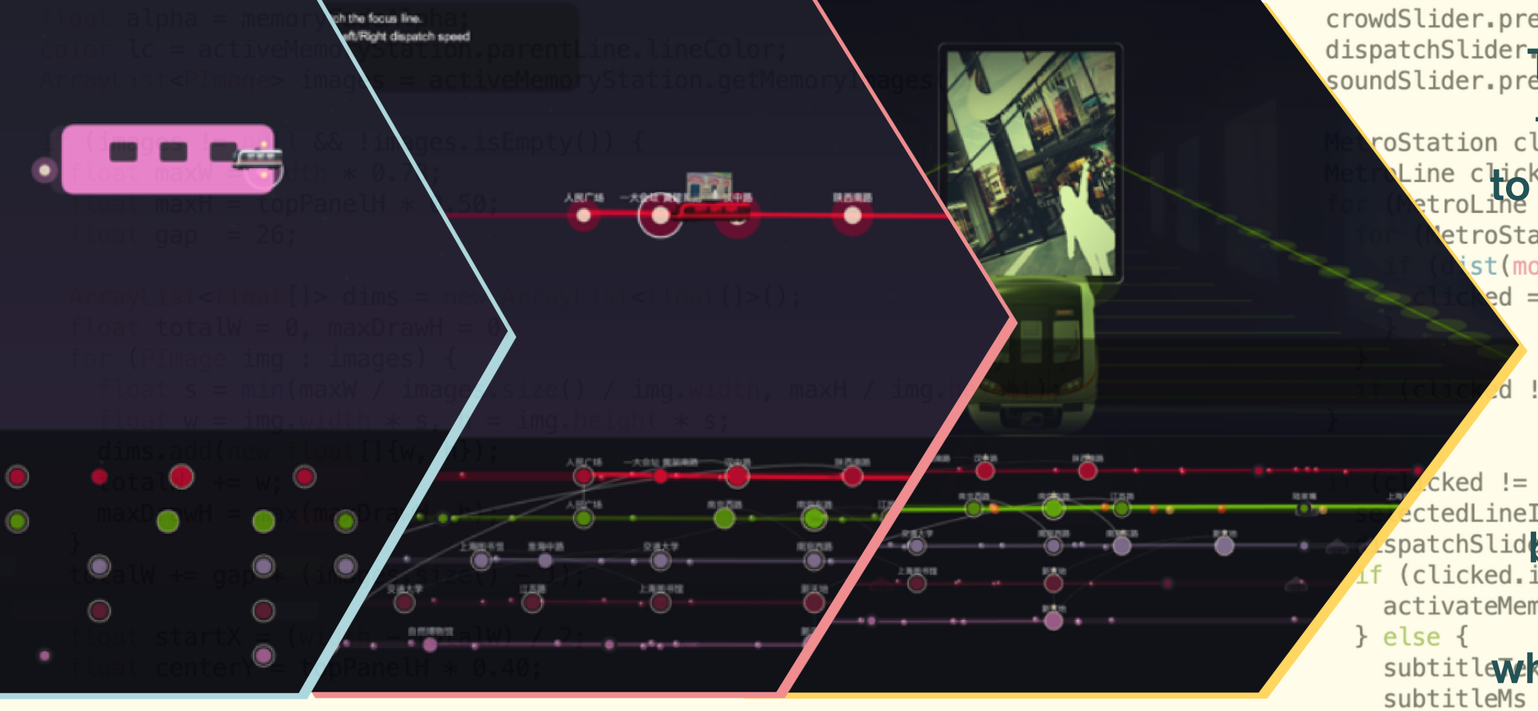
  int bars = int(map(energyN, 0, 1, 5, 16
  float barW = w / (bars * 1.8);
  float startX = left + (w - bars * barW

  for (int i = 0; i < bars; i++) {
    float val = spectrumVals[i % spectrum
    float barH = map(val, 0.3, 1, h * 0.1
    float pulse = sin(frameCount * 0.1 +
    barH *= (1 + pulse);

    float bx = startX + i * barW * 1.4;
    float by = yTop + h - barH;

    noStroke();

    if (era.equals("2020s")) {
      fill(red(glowCol), green(glowCol),
      rect(bx, by, barW, barH, 3);
      fill(255, 255, 255, 60);
      rect(bx - 1, by - 1, barW + 2, barH
    } else if (era.equals("2010s")) {
      fill(255, 255, 255, 200);
      rect(bx, by, barW, barH, 5);
    } else {
      fill(0, 0, 0, 120);
      rect(bx, by, barW, barH);
      fill(255, 255, 255, 200);
```



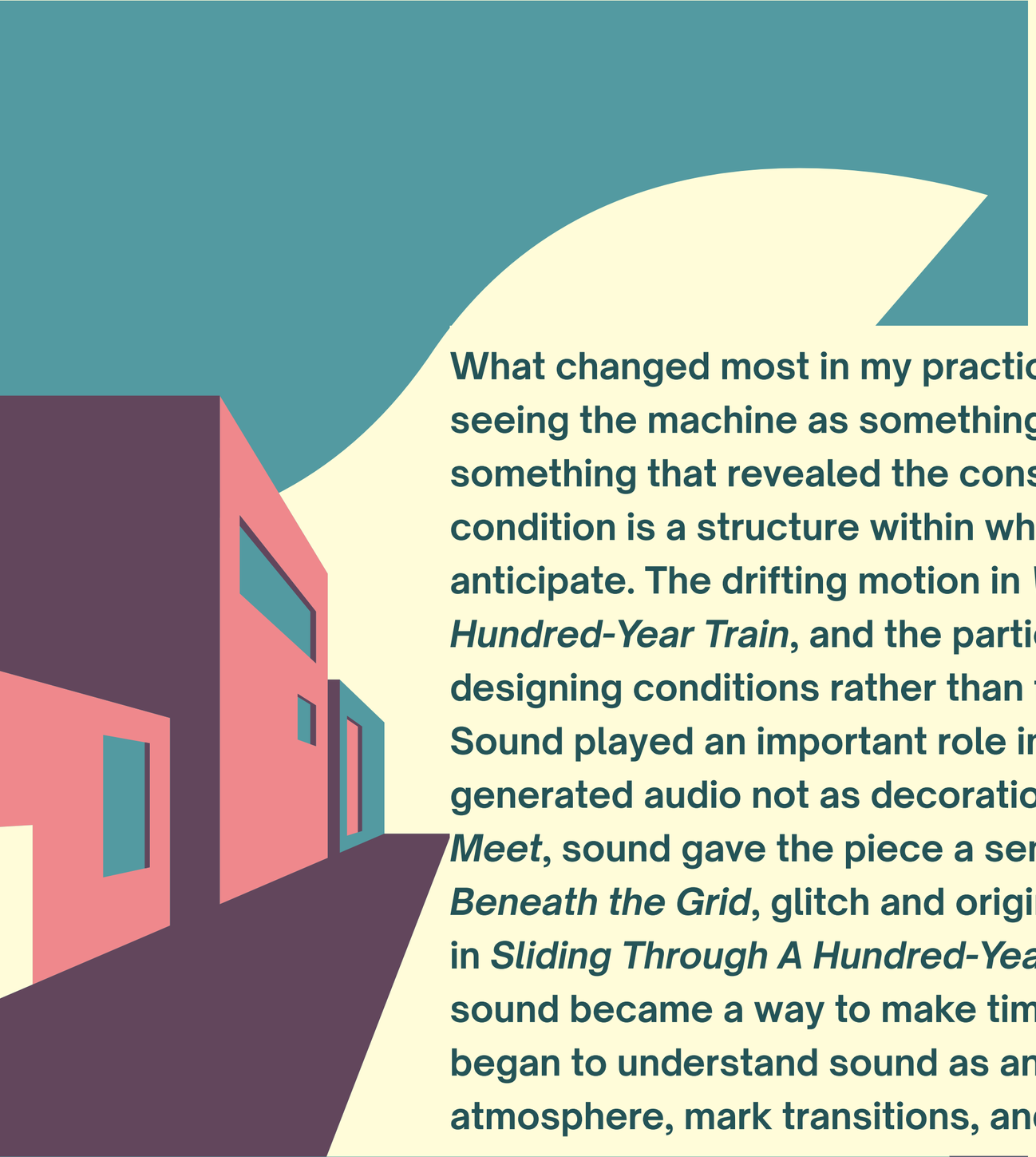
This was the project where I felt the strongest shift from control toward collaboration. The system had to be tuned carefully, and a lot of the final form came from solving things that were not working: audio overlap, drawing order, video loading, and memory triggers. What surprised me most is that these technical problems were not separate from the concept; they actually helped me understand it better. In the end, the project felt less like a map of Shanghai Metro and more like a way of listening to what a city remembers but does not officially record.

Conducting the Underground Symphony goes further by embedding private experience inside public infrastructure. The Shanghai metro runs generatively: trains follow fixed station sequences, pentatonic tones sound at arrivals, and passenger agents cluster around musical pulses in an energy-based simulation that can stabilize or spiral depending on crowd density. But certain stations — the site of the Expo, a theater on weekends, the stop for a first internship — carry personal photographs. When a train arrives at one of these stations, memory surfaces over the generative layer: a faded image with a soft line-color tint, a caption, a chord rather than a single note. The city does not remember these things. The metro map carries no trace of them. I embedded them in the rules of a simulation, and the simulation makes them visible at intervals determined by its own logic — not by my intention, but by the system's unfolding. That act of encoding private experience inside a public computational system is itself a formal claim about what data can hold, and about the invisible archive of feeling that underlies every infrastructure we move through.



IV. THE SPACE BETWEEN

Looking across six projects, I notice a pattern. I always began with a city and arrived somewhere more personal. The city is legible. You can map it, photograph it, date it. But what interested me was what it does not record: the feeling of walking the same campus in four different emotional states, the memory of an afternoon at a metro station that appears on no official map, the minute in which four lines of code briefly shared a canvas and produced something I did not plan. Code became the instrument for holding both the systemic and the intimate simultaneously. I did not set out to make personal work. I set out to make work about cities. But every time I built a system, I ended up putting something of my own inside it.



What changed most in my practice was the way I thought about authorship. I stopped seeing the machine as something that simply executed my ideas and started seeing it as something that revealed the consequences of my rules. An effect is a result I aimed for. A condition is a structure within which results become possible, including results I did not anticipate. The drifting motion in *When We Meet*, the shifting states in *Sliding Through A Hundred-Year Train*, and the particle behavior in *City Through Four Lenses* all came from designing conditions rather than fixed outcomes.

Sound played an important role in this process too. In several projects, I used music and generated audio not as decoration, but as part of the system's behavior: in *When We Meet*, sound gave the piece a sense of duration and performance; in *The Universe Beneath the Grid*, glitch and original music helped separate the two modes emotionally; in *Sliding Through A Hundred-Year Train* and *Conducting the Underground Symphony*, sound became a way to make time, history, and movement feel physical. Over time, I began to understand sound as another kind of structure — one that could carry atmosphere, mark transitions, and make the system feel alive.

What I learned this semester is that generative art is not about giving up control, but about designing a system carefully enough that it can surprise you. The final output matters, but so do the rules, the failed versions, the adjustments, and the moments when the work behaves in a way I did not expect.

That is what I now think code is for: not just producing images, but creating a conversation between intention and emergence.

In my work, the city became the subject, the system became the method, and the space between them — where image, sound, and behavior meet — became the artwork.

THANK YOU

Yi Ru (Raya)

Final Portfolio · COMPSCI 208

Acknowledgments: Prof. Ziv Ze'ev Cohen

